Richard Prange

Charlotte, NC | rprange@charlotte.edu | github.com/richard | linkedin.com/in/richard | home/richard

Education

Western Carolina University — Cullowhee, NC
Bachelor of Science in Computer Science

Western Carolina University — Cullowhee, NC
Bachelor of Science in Mathematics

University of North Carolina at Charlotte — Charlottee, NC
Masters of Science in Mathematics — *Aug, 2025 → Current*

Relevant Courses: *Data Structures and Algorithms, Software Development & Engineering, Problem Solving and Programming 1 & 2 (OOP), Cloud Computing, Information Security, Computer Networking, Operating Systems*

Experience

Mathematics and Computer Science Tutor — Jan, 2022 → May, 2025
Western Carolina University — *Cullowhee, NC*

- Provide personalized assistance to students in understanding complex mathematical and computational concepts
- Foster critical thinking and problem solving skills through active learning techniques

Projects

WheeBFit | *React, TypeScript, Node.js, Express, JWT, Heroku, SQL* — May, 2025

- Developed and deployed a full stack web application for a client
- Implemented a highly interactive interface designed to keep students entertained while allowing teachers to track, display, and monitor a class' progress
- Information is stored in a SQL database and is accessible to authorized users
- **Visit Here:** https://wheebfit.com

ChessMeister | *Java, JavaFX* — May, 2024

- Developed a chess program that allows 2-4 players to play chess through a CLI or GUI
- Implemented valid check, and checkmate algorithms, used to end the game or validate moves
- Incorporated functionality to allow for users to undo, and redo moves
- Designed a loosely coupled system using interfaces and model view controller to allow for seamless future expansion on the user interface
- **Git Repo:** https://gitfront.io/r/ricwp12/6W3trXvVkj6g/ChessMeister/

Rusty Battleship | *Rust* — Jan, 2025

- Developed a program that allows N-players to connect to a server and play a CLI version of BattleShip
- Provided functionality using threads that allow users to continously recieve updates from the server while issuing commands via. the command line to view the game state or make a move
- Implemented shared memory to allow the sevrer to spawn threads for each user which efficiently managed user requests, messages, and state updates
- **Git Repo:** https://gitfront.io/r/ricwp12/U24fPLcj6QfM/Battleship/

GraphAnalyzer | *Java* — Jan, 2024

- Developed a program that coverts .txt files to undirected graphs and allows users to to perform operations on the graph.
- Implemented alogirthms to perform **Depth First Search, Breadth First Search,** and **Transitive Closure**
- **Git Repo:** https://gitfront.io/r/ricwp12/U24fPLcj6QfM/Battleship/

Technical Skills

Languages: Python, Java, R, C, JavaScript, Rust, SQL, TypeScript, C^{++}
Frameworks/Tools: Node.js, Express.js, TailwindCSS, SFML, React **Other**: Linux, Docker, Heroku, Google Cloud Platform, Git